Low-Level Design – Dashboard

TaskFlow – Intelligent Task Allocation & Tracking System

> **Document:** Low-Level Design (sub-module 1 – DASH) · **Version:** 1.0 · **Status:** Draft
> **Project type:** Personal Project (Skill Development) · **Date:** 31-Oct-2026
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> **Based on:** HLD – Dashboard v1.0, FRD – Dashboard v1.0 (FR-DASH-01 – FR-DASH-75), TaskFlow code as built
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Document Control

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version: Overview (route selection, counter endpoints) and Analytics (Administrator, Manager, Employee analytics, filters, drill-down rules), derived from the code and verified against the code fact sheet

Related documents: BRD – Dashboard (`../../../BRD/01-Dashboard/01.1-Dashboard/BRD-Dashboard.md`) · FRD – Dashboard (`../../../FRD/01-Dashboard/01.1-Dashboard/FRD-Dashboard.md`) · HLD – Dashboard (`../../../HLD/01-Dashboard/01.1-Dashboard/HLD-Dashboard.md`) · LLD – Overall (`../../00-Overall/LLD-TaskFlow-Overall.md`) · LLD – Task Management, drill-down target (`../../02-TaskFlow-Monitoring/02.1-Task-Management/LLD-Task-Management.md`) · TDD (`TF - Documents/TF - TDD/TDD-TaskFlow.md` v2.0)

1. Introduction

1.1 Purpose

Detailed design of the Dashboard sub-module: every controller action, the classes and files behind it, the DTOs, the filter validation, the scope and bucket rules, the repository queries, the tables it reads, the tests, and the traceability from FR-DASH-nn to endpoint, class and test. It is detailed enough to maintain the code without reading it first.

1.2 Scope & Conventions

- Child modules: **Overview** (route selection, guards, the three page components, the counter endpoints of `DashboardController`) and **Analytics** (`AnalyticsController`, the three analytics services, the scopes, the shared bucket rules and the filter bar / chart components).
- **Endpoints in this sub-module: 8** (facts.md §3: `DashboardController` 3 + `AnalyticsController` 5). All are `GET`; the sub-module writes nothing.

- Naming: `*Controller`, `I*Service` / `*Service`, `I*Repository` / `*Repository`, DTOs `*Dto`, query records `AnalyticsFilter`; scopes `*DashboardScope`; shared rules `*Rules` in `Common/Rules`.
- API base path: `/api/v1/dashboard`; JSON camelCase with enums as names (`JsonStringEnumConverter`); errors as `ProblemDetails` (`application/problem+json`).
- Paths below are relative to `backend/src/TaskFlow.Api/` (backend) and `frontend/src/app/` (frontend).

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2. Code Structure

2.1 Backend Folder Layout

```
TaskFlow.Api/
  Controllers/Dashboard/                 DashboardController.cs, AnalyticsController.cs
  Services/Dashboard/Overview/           IDashboardService.cs, DashboardService.cs
  Services/Dashboard/Analytics/Admin/    IAdminAnalyticsService.cs, AdminAnalyticsService.cs,
AnalyticsBuckets.cs
  Services/Dashboard/Analytics/Manager/  IManagerAnalyticsService.cs,
ManagerAnalyticsService.cs
  Services/Dashboard/Analytics/Employee/ IEmployeeAnalyticsService.cs,
EmployeeAnalyticsService.cs
  Services/Dashboard/Analytics/Scopes/   IDashboardScope.cs, AdminDashboardScope.cs,
IManagerDashboardScope.cs,
                                         ManagerDashboardScope.cs, IEmployeeDashboardScope.cs,
EmployeeDashboardScope.cs,
                                         AnalyticsFilterResolver.cs,
ResolvedAnalyticsFilter.cs, ScopedTasks.cs
  Repositories/Dashboard/Overview/       IDashboardRepository.cs, DashboardRepository.cs,
DashboardTaskCounts.cs,
                                         DashboardDayBounds.cs, DashboardTaskScope.cs
  Repositories/Dashboard/Analytics/      IAnalyticsRepository.cs, AnalyticsRepository.cs,
AnalyticsCounting.cs,
                                         AnalyticsTaskSet.cs, AnalyticsDateRange.cs,
DashboardTaskFilters.cs,
                                         OverdueTaskRow.cs, DeadlineTaskRow.cs,
ProjectProgressRow.cs, QuarterHourCount.cs
  Dtos/Dashboard/                        DashboardDto.cs
  Dtos/Dashboard/Analytics/              AnalyticsFilter.cs, AnalyticsDtos.cs,
ManagerAnalyticsDtos.cs, EmployeeAnalyticsDtos.cs
  Common/Rules/                          TaskBucketRules.cs, DeadlineBucketRules.cs,
ManagerTeamThresholds.cs (shared)
  Common/Exceptions/                     AnalyticsExceptions.cs
  Config/                                ServiceRegistration.cs, RepositoryRegistration.cs (DI,
all scoped)
```

There is no FluentValidation validator in this sub-module: the analytics filter is validated in `AnalyticsFilterResolver` and the employee period in `EmployeeAnalyticsService.ParsePeriod`.

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2.2 Frontend Folder Layout

```
src/app/modules/dashboard/
  dashboard.routes.ts                    DASHBOARD_ROUTES: admin, manager, me
  pages/                                 admin-dashboard, manager-dashboard, employee-dashboard
(.ts/.html)
  analytics/components/                  task-analytics (+ .model.ts), department-analytics,
employee-analytics,
                                         manager-analytics, project-analytics, team-attention,
team-task-analytics,
                                         team-employee-analytics
  analytics/services/                    dashboard-analytics.service.ts, manager-
analytics.service.ts,
                                         employee-analytics.service.ts (each + .model.ts)
  analytics/models/analytics.model.ts    interfaces, ANALYTICS_BASES, EMPLOYEE_PERIODS
  components/                            dashboard-filter-bar.component.ts/.html
  components/charts/                     kpi-card, donut-chart, stacked-bar-chart, line-chart,
progress-list (+ .model.ts)
  services/dashboard.service.ts          DashboardService (counter endpoint, not injected
anywhere)
  models/dashboard.model.ts              EmployeeDashboardStats
  utils/                                 chart-colors.ts, chart-providers.ts
src/app/shared/utils/                    analytics-filter.ts, drill-down.ts
src/app/shared/auth/                     role-routes.ts (dashboardRouteFor), permissions.ts,
auth.guard.ts (permissionGuard)
src/app/layouts/nav-items.ts             single "Dashboard" menu item
```

2.3 Layer Rules

Rule	Enforced by
Controller → Service → Repository; controllers hold no query logic	Code structure; shared architecture rules in the overall LLD
Scope (which tasks) is decided only by a `*DashboardScope`; services never build task filters themselves	`IDashboardScope`, `IManagerDashboardScope`, `IEmployeeDashboardScope`
One bucket definition for every count and for the Task List drill-down	`TaskBucketRules`, `DeadlineBucketRules` (also used by the Task Management repository)
Read only: no `SaveChanges`, no notification, mail or activity entry	`AsNoTracking` queries; no `IUnitOfWork` dependency in this sub-module
Frontend pages call only their own analytics service; the services own the HTTP calls and state	`DashboardAnalyticsService`, `ManagerAnalyticsService`, `EmployeeAnalyticsService`

3. Cross-Cutting Design

Shared cross-cutting design (JWT authentication, `perm:` policies, `ValidationFilter`, `GlobalExceptionHandler`, `DeniedActionAuditHandler`, logging, DI) is described once in the overall LLD (`../../00-Overall/LLD-TaskFlow-Overall.md`). Dashboard-specific points:

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Concern	Design
Validation	No FluentValidation validator. `AnalyticsFilterResolver.ResolveAsync` checks range, maximum span, date basis and (Administrator) that named ids exist; `EmployeeAnalyticsService.ParsePeriod` checks the period. The Angular `validateRange` stops an empty or reversed range before a request is sent.
Error handling	Analytics exceptions derive from `AppException` types and are turned into `ProblemDetails` by `GlobalExceptionHandler` (table 3.1). The Angular services ignore the server message and show MSG-DASH-01 / MSG-DASH-02.
Transactions	None – read only.
Authorization	Each analytics action needs **both** a `perm:` policy (`dashboard.analytics.view` or `dashboard.filters.use`) and a `RequirePermission` view key; scopes enforce data visibility (Organisation / Team / Own).
Logging	Refused policy checks are logged by `DeniedActionAuditHandler` (user id, method, path, policy).
Time & time zone	Tasks are stored in UTC. Day boundaries are computed in the application timezone (`IAppClock.GetTimeZoneAsync`, fallback `Application:Timezone`) with `ReminderCalculator.LocalDate` / `LocalMidnightUtc` and converted to UTC instants; ranges are end-exclusive (`ToUtcExclusive`). "Now" is `IAppClock.UtcNow`.
Caching	`ManagerDashboardScope` caches team ids and project ids per manager for the lifetime of one request (scoped service).

3.1 Exception → HTTP Status

Exception	Base class	Status	When
`AnalyticsFilterException`	`BusinessRuleException`	400	End before start, span ≥ 3660 days, undefined date basis, unknown employee period
`AnalyticsForbiddenException`	`ForbiddenException`	403	Manager filter names an employee outside the team or a project outside the manager's scope
`AnalyticsEntityNotFoundException`	`NotFoundException`	404	Named department, project, employee or manager does not exist
Policy failure (`perm:` or `RequirePermission`)	–	403	Missing `dashboard.analytics.view`, `dashboard.filters.use`, `dashboard.view.organization` or `dashboard.view.team`
Caller id claim not an integer	– (controller returns `Unauthorized()`)	401	Manager / employee endpoints
Any other exception	–	500	"Something went wrong. Please try again."

4. Module Designs

4.1 Module: Dashboard (DASH)

4.1.1 Child module: Overview

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Route selection, route guards, the menu item and the three page components (frontend), plus the counter endpoints of `DashboardController` (backend). The counter endpoints exist and are tested but no Angular screen calls them: the cards on the three pages are loaded from the Analytics endpoints (4.1.2).

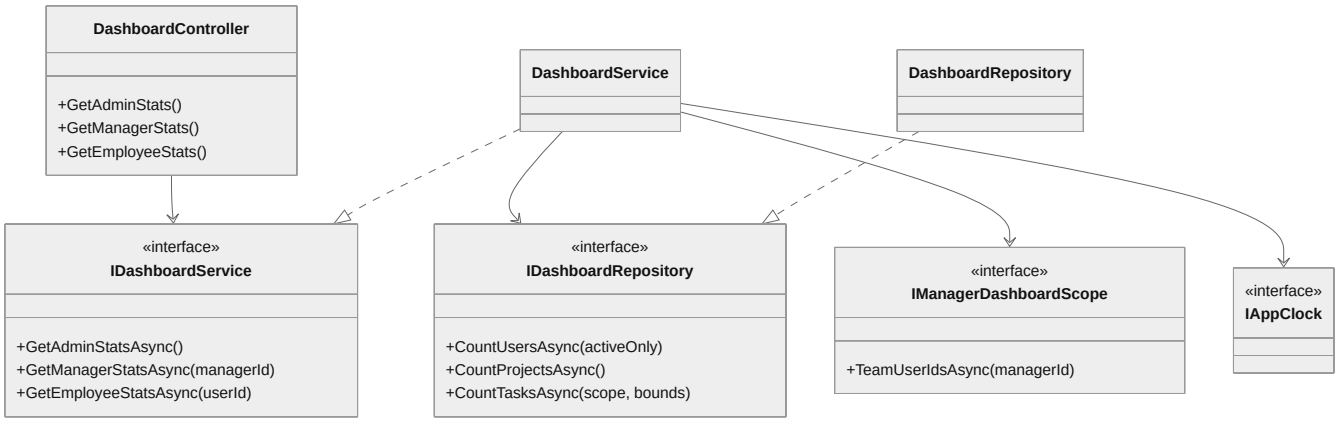
Class / file inventory

Layer	Type	File	Responsibility
Controller	`DashboardController`	`Controllers/Dashboard/DashboardController.cs`	Three counter actions; reads the caller id with `TryGetCallerId`
Service	`IDashboardService` / `DashboardService`	`Services/Dashboard/Overview/`	Builds day bounds; calls the repository with an Organisation, Team or Own scope
Repository	`IDashboardRepository` / `DashboardRepository`	`Repositories/Dashboard/Overview/`	`CountUsersAsync`, `CountProjectsAsync`, `CountTasksAsync` (one grouped query)
Records	`DashboardTaskScope`, `DashboardDayBounds`, `DashboardTaskCounts`	`Repositories/Dashboard/Overview/`	Scope (team ids / own id), local-day bounds as UTC, the seven counters
DTOs	`AdminDashboardDto`, `ManagerDashboardDto`, `EmployeeDashboardDto`	`Dtos/Dashboard/DashboardDto.cs`	Response contracts
Dependency	`IManagerDashboardScope.TeamUserIdsAsync`	`Services/Dashboard/Analytics/Scopes/`	Manager's active team (owned by Analytics)
Dependency	`IAppClock`	`Services/Settings/ApplicationSettings/Clock/`	`UtcNow`, application timezone
Validator	–	–	No input
Frontend routes	`DASHBOARD_ROUTES`	`modules/dashboard/dashboard.routes.ts`	`admin` (`permissionGuard`, `dashboard.view.organization`), `manager` (`permissionGuard`, `dashboard.view.team`), `me` (no guard)
Frontend landing	`dashboardRouteFor`, `ROLE_ROUTES`, `DEFAULT_ROUTE`	`shared/auth/role-routes.ts`	Administrator → `/admin`, Manager → `/manager`, else by permission, else `/me`
Frontend menu	nav items	`layouts/nav-items.ts`	Single **Dashboard** item routed with `dashboardRouteFor(...)`
Frontend pages	`AdminDashboardComponent`, `ManagerDashboardComponent`, `EmployeeDashboardComponent`	`modules/dashboard/pages/`	The three screens (`/admin`, `/manager`, `/me`); inject the analytics services

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Layer	Type	File	Responsibility
Frontend service	`DashboardService`	`modules/dashboard/services/dashboard.service.ts`	`getEmployeeStats()` → counter endpoint; not injected anywhere

Class diagram



API contracts (route prefix `[Route("api/v1/dashboard")]`)

#	Method	Route	Permission / policy	Request	Response	Status codes
1	GET	`/api/v1/dashboard/admin`	`RequirePermission(dashboard.view.organization)`	–	`AdminDashboardDto`	200, 401, 403
2	GET	`/api/v1/dashboard/manager`	`RequirePermission(dashboard.view.team)` (Administrator passes by scope expansion)	– (caller id from `ClaimTypes.NameIdentifier`)	`ManagerDashboardDto`	200, 401, 403
3	GET	`/api/v1/dashboard/employee`	`[Authorize]` only (`dashboard.view.own` is Always On and not stored)	– (caller id from claim)	`EmployeeDashboardDto`	200, 401

Request / response example

```
{ "GET /api/v1/dashboard/manager": { "totalTasks": 64, "pendingTasks": 22, "tasksDueToday": 3,
  "upcomingDeadlines": 9, "overdueTasks": 4, "completedTasks": 40 } }
```

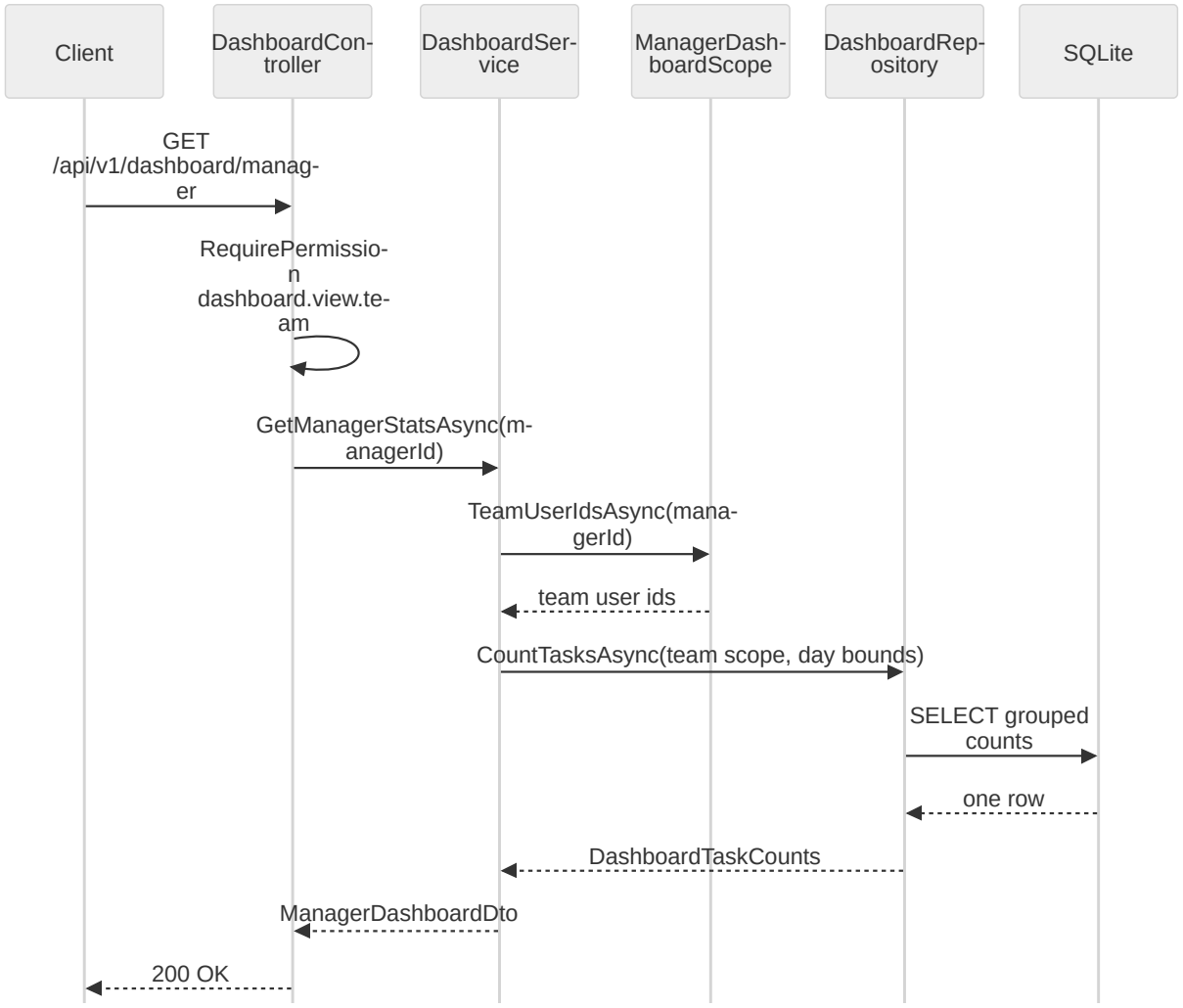
Service logic – `DashboardService`

1. `BoundsAsync`: read the timezone and `UtcNow` from `IAppClock`; `today` = `ReminderCalculator.LocalDate`; build `DashboardDayBounds(now, todayStart, tomorrowStart, upcomingHorizon)` where the horizon is the local midnight of today + `UpcomingDeadlineWindowDays` (7) + 1.
2. `GetAdminStatsAsync`: `CountUsersAsync(false)`, `CountUsersAsync(true)`, `CountProjectsAsync()`, then `CountTasksAsync(new DashboardTaskScope(), bounds)`; map to `AdminDashboardDto` with `PendingTasks` = the **open** count.

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3. `GetManagerStatsAsync(managerId)`: team = `IManagerDashboardScope.TeamUserIdsAsync(managerId)`; `CountTasksAsync(new DashboardTaskScope(TeamUserIds: team), bounds)`; map to `ManagerDashboardDto`.
4. `GetEmployeeStatsAsync(userId)`: `CountTasksAsync(new DashboardTaskScope(OwnUserId: userId), bounds)`; map to `EmployeeDashboardDto`.
5. Nothing is persisted; no side effects.

Sequence diagram



Repository methods

Method	Query / purpose	Notes
`CountUsersAsync(activeOnly)`	`Users.CountAsync(!activeOnly || IsActive)`	Soft-delete query filter hides deleted users
`CountProjectsAsync()`	`Projects.CountAsync()`	Soft-delete query filter

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Method	Query / purpose	Notes
`CountTasksAsync(scope, bounds)`	`AsNoTracking`, `!IsDeleted`; team / own = assignee **or** a `TaskRecipients` row; one `GroupBy(_ => 1)` producing Total, Open, DueToday, Upcoming, Overdue, InProgress, Completed	Open = not Completed and not Cancelled; DueToday / Upcoming / Overdue count open tasks only; InProgress = InProgress or InQA; zeros when no rows. Uses `IX_Tasks_AssignedToUserId`, `IX_TaskRecipients_TaskId_UserId`, `IX_Tasks_Deadline`

Frontend logic – route selection (FR-DASH-01 – FR-DASH-05)

1. `dashboardRouteFor(role, permissions)`: `ROLE_ROUTES[role]` (Administrator → `/admin`, Manager → `/manager`) wins; else `dashboard.view.organization` → `/admin`; else `dashboard.view.team` → `/manager`; else `DEFAULT_ROUTE` `/me`.
2. Sign-in and Security Setup navigate to this route; the **Dashboard** menu item uses the same function.
3. `permissionGuard` on `admin` / `manager` checks `data.permissions`; when the key is missing it redirects to `dashboardRouteFor(...)` without a message. `me` has no guard (the shell `authGuard` + `mfaSetupGuard` apply).

Validation rules

Field	Rule	Message
Caller id (`NameIdentifier` claim)	Must parse as `int` (manager, employee)	– (401)

Error cases

Case	Exception	Status
Caller id missing / not numeric	– (`Unauthorized()`)	401
Missing `dashboard.view.organization` / `dashboard.view.team`	Policy	403

State machine: *Not applicable for this child module.*

4.1.2 Child module: Analytics

Every figure on the three dashboard pages: summary cards, donuts, trend, deadline buckets, department / employee / manager / project breakdowns, the Manager attention lists and workload flags, the Employee deadlines, projects and department overview, plus the filter options.

Class / file inventory

Layer	Type	File	Responsibility
Controller	`AnalyticsController`	`Controllers/Dashboard/AnalyticsController.cs`	Five actions; policy + view-key attributes; caller id for manager / employee

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Layer	Type	File	Responsibility
Service	`IAdminAnalyticsService` / `AdminAnalyticsService`	`Services/Dashboard/Analytics/Admin/`	Organisation analytics and filter options
Service	`IManagerAnalyticsService` / `ManagerAnalyticsService`	`Services/Dashboard/Analytics/Manager/`	Team analytics, flags, attention lists, team filter options
Service	`IEmployeeAnalyticsService` / `EmployeeAnalyticsService`	`Services/Dashboard/Analytics/Employee/`	Own analytics, period, deadlines, projects, department overview
Helper	`AnalyticsBuckets` (internal static)	`Services/Dashboard/Analytics/Admin/AnalyticsBuckets.cs`	`Empty`, `Sum` of `BucketCountsDto`
Scope	`IDashboardScope` / `AdminDashboardScope`	`Services/Dashboard/Analytics/Scopes/`	Resolve with entity checks; build `AnalyticsTaskSet` with department / project / employee / manager
Scope	`IManagerDashboardScope` / `ManagerDashboardScope`	`Services/Dashboard/Analytics/Scopes/`	Drop department / manager; `EnsureInScopeAsync`; team and project ids (cached per request)
Scope	`IEmployeeDashboardScope` / `EmployeeDashboardScope`	`Services/Dashboard/Analytics/Scopes/`	`OwnTasks(userId)` = `AnalyticsTaskSet(OwnUserId: userId)`
Resolver	`AnalyticsFilterResolver`, `ResolvedAnalyticsFilter`, `ScopedTasks`	`Services/Dashboard/Analytics/Scopes/`	Default range, limits, basis, UTC bounds; resolved filter; set with and without dates
Repository	`IAnalyticsRepository` / `AnalyticsRepository`	`Repositories/Dashboard/Analytics/`	38 query methods (counts, groupings, lists, lookups)
Repository helpers	`AnalyticsCounting` (internal), `DashboardTaskFilters`	`Repositories/Dashboard/Analytics/`	Bucket totals and per-key counts; `ApplyDateRange`, `ApplyEntityFilters`, `TeamUserIds`

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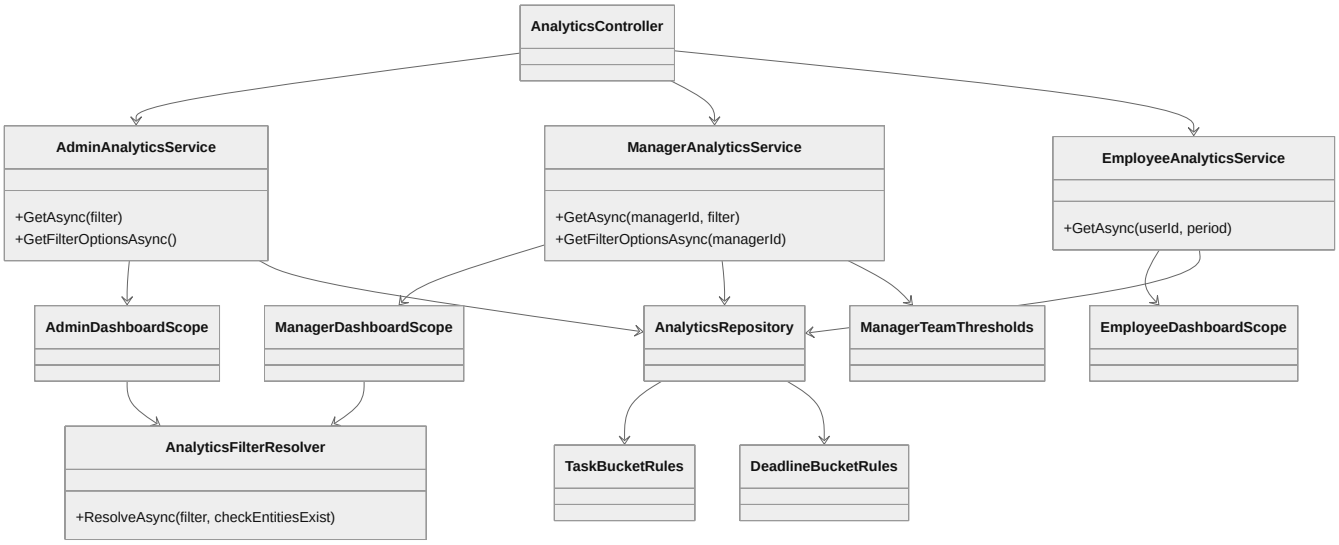
Layer	Type	File	Responsibility
Records	`AnalyticsTaskSet`, `AnalyticsDateRange`, `OverdueTaskRow`, `DeadlineTaskRow`, `ProjectProgressRow`, `QuarterHourCount`	`Repositories/Dashboard/Analytics/`	Query inputs and row shapes
Shared rules	`TaskBucketRules` (+ `TaskBucket`), `DeadlineBucketRules` (+ `DeadlineBucket`, `DeadlineBounds`), `ManagerTeamThresholds`	`Common/Rules/`	Bucket, deadline window, workload flag rules
Exceptions	`AnalyticsFilterException`, `AnalyticsEntityNotFoundException`, `AnalyticsForbiddenException`	`Common/Exceptions/AnalyticsExceptions.cs`	400 / 404 / 403
DTOs	`AnalyticsFilter`, `AnalyticsDateBasis`	`Dtos/Dashboard/Analytics/AnalyticsFilter.cs`	Query input
DTOs	`BucketCountsDto`, `AppliedFilterDto`, `AnalyticsSummaryDto`, `PriorityCountDto`, `DeadlineBucketsDto`, `TrendPointDto`, `TrendDto`, `DepartmentAnalyticsDto`, `EmployeeAnalyticsDto`, `EmployeeAnalyticsSectionDto`, `ManagerAnalyticsDto`, `ProjectAnalyticsDto`, `AdminAnalyticsDto`, `AnalyticsOptionDto`, `AnalyticsFilterOptionsDto`	`Dtos/Dashboard/Analytics/AnalyticsDtos.cs`	Administrator contracts (shared parts reused by Manager / Employee)
DTOs	`TeamSummaryDto`, `TeamEmployeeDto`, `DelayedTaskDto`, `ProjectAttentionDto`, `TeamAttentionDto`, `ManagerAnalyticsResponseDto`, `ManagerFilterOptionsDto`	`Dtos/Dashboard/Analytics/ManagerAnalyticsDtos.cs`	Manager contracts
DTOs	`EmployeeAnalyticsPeriod`, `EmployeeDepartmentDto`, `EmployeeSummaryDto`, `EmployeeDeadlineDto`, `EmployeeProjectDto`, `EmployeeAnalyticsResponseDto`	`Dtos/Dashboard/Analytics/EmployeeAnalyticsDtos.cs`	Employee contracts
Validator	–	–	Rules live in `AnalyticsFilterResolver` and `ParsePeriod`

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Layer	Type	File	Responsibility
Frontend services	`DashboardAnalyticsService`, `ManagerAnalyticsService`, `EmployeeAnalyticsService`	`modules/dashboard/analytics/services/`	Signal state (`filter`, `options`, `data`, `loading`, `error`), `start`, `update`, `reset`, `refresh`, `clear` (`setAllEmployees` Admin, `setPeriod` Employee); `switchMap` drops stale responses
Frontend filter bar	`DashboardFilterBarComponent` (`tf-dashboard-filter-bar`)	`modules/dashboard/components/`	Preset chips, From / To, basis, Department / Project / Manager / Employee
Frontend analytics components	`TaskAnalyticsComponent`, `DepartmentAnalyticsComponent`, `EmployeeAnalyticsComponent`, `ManagerAnalyticsComponent`, `ProjectAnalyticsComponent`, `TeamAttentionComponent`, `TeamTaskAnalyticsComponent`, `TeamEmployeeAnalyticsComponent`	`modules/dashboard/analytics/components/`	Sections of the Administrator and Manager pages
Frontend charts	`KpiCardComponent`, `DonutChartComponent`, `StackedBarChartComponent`, `LineChartComponent`, `ProgressListComponent`; `provideDashboardCharts`	`modules/dashboard/components/charts/`, `modules/dashboard/utils/`	Chart.js (`ng2-charts`) wrappers; colours in `chart-colors.ts`
Frontend utils	`RANGE_PRESETS`, `BASIS_LABELS`, `presetRange`, `defaultFilter`, `detectPreset`, `validateRange`, `toQueryParams`; `drillDownQuery`, `employeeDrillQuery`, `parseDrillDown`, `DRILL_BUCKETS`, `DRILL_DEADLINE_BUCKETS`, `NO_GROUP_ID`	`shared/utils/analytics-filter.ts`, `shared/utils/drill-down.ts`	Presets, client range check, query params, drill-down query builder

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Class diagram



The services depend on the interfaces (`IAdminAnalyticsService`, `IDashboardScope`, `IAnalyticsRepository`, …); all are registered as scoped in `Config/ServiceRegistration.cs` and `Config/RepositoryRegistration.cs`.

API contracts (route prefix `[Route("api/v1/dashboard")]`; both the `perm:` policy and the `RequirePermission` key must pass)

#	Method	Route	Permission / policy	Request	Response	Status codes
4	GET	`/api/v1/dashboard/admin/analytics`	`perm:dashboard.analytics.view` + `RequirePermission(dashboard.view.organization)`	`AnalyticsFilter` (query)	`AdminAnalyticsDto`	200, 400, 401, 403, 404
5	GET	`/api/v1/dashboard/admin/analytics/filters`	`perm:dashboard.filters.use` + `RequirePermission(dashboard.view.organization)`	–	`AnalyticsFilterOptionsDto`	200, 401, 403
6	GET	`/api/v1/dashboard/manager/analytics`	`perm:dashboard.analytics.view` + `RequirePermission(dashboard.view.team)`	`AnalyticsFilter` (query; `departmentId`, `managerId` ignored)	`ManagerAnalyticsResponseDto`	200, 400, 401, 403, 404
7	GET	`/api/v1/dashboard/manager/analytics/filters`	`perm:dashboard.filters.use` + `RequirePermission(dashboard.view.team)`	–	`ManagerFilterOptionsDto`	200, 401, 403
8	GET	`/api/v1/dashboard/employee/analytics`	`perm:dashboard.analytics.view` (plus a plain `[Authorize]`)	`period` query: `7d`, `30d` (default), `month`	`EmployeeAnalyticsResponseDto`	200, 400, 401, 403

`AnalyticsFilter` query parameters: `from`, `to` (`DateOnly`, `yyyy-MM-dd`), `basis` (`Created` default, `Deadline`, `Completed`), `departmentId`, `projectId`, `employeeId`, `managerId` (int, optional; `0` means "No

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department" / "No project" for drill-down), `allEmployees` (bool, Administrator only – every employee instead of the top 10).

Request / response example

```
{
  "request": "GET /api/v1/dashboard/manager/analytics?from=2026-10-02&to=2026-10-
31&basis=Created",
  "response": {
    "filter": { "from": "2026-10-02", "to": "2026-10-31", "basis": "Created", "departmentId":
null,
                "projectId": null, "employeeId": null, "managerId": null },
    "summary": { "teamSize": 8, "totalProjects": 2, "totalTasks": 22, "completionPercent": 55,
                 "tasks": { "pending": 5, "inProgress": 3, "completed": 12, "overdue": 2,
"total": 22, "open": 10, "completionPercent": 55 } },
    "byEmployee": [ { "userId": 7, "name": "Asha Nair", "highWorkload": true, "hasOverdue":
true,
                      "tasks": { "pending": 3, "inProgress": 2, "completed": 1, "overdue": 1,
"total": 7, "open": 6, "completionPercent": 14 } } ],
    "attention": {
      "delayedTasks": [ { "taskId": 91, "taskCode": "TSK-0091", "title": "Fix login",
"assigneeName": "Asha Nair",
                          "projectName": "Portal", "daysOverdue": 3 } ],
      "projectsNeedingAttention": [ { "projectId": 3, "name": "Portal", "overdueTasks": 2 } ]
    }
  }
}
```

```
{ "status": 400, "detail": "The end date cannot be before the start date." }
```

Values are illustrative; `tasksByStatus`, `tasksByPriority` and `byProject` are omitted for brevity. The Administrator response (`AdminAnalyticsDto`) adds `deadlineBuckets`, `trend`, `byDepartment`, `byEmployee` (`items`, `totalEmployees`), `byManager` and `byProject`; the Employee response (`EmployeeAnalyticsResponseDto`) has `period`, `from`, `to`, `department`, `summary`, `tasksByStatus`, `tasksByPriority`, `upcomingDeadlines`, `projects`, `trend`, `completedInPeriod`.

Service logic – `AnalyticsFilterResolver.ResolveAsync(filter, checkEntitiesExist)` (Administrator and Manager)

1. Read timezone and `UtcNow` from `IAppClock`; `today` = local date.
2. `to` = `filter.To ?? today`; `from` = `filter.From ?? to - 29 days` (`DefaultRangeDays = 30`).
3. `from > to` → `AnalyticsFilterException("The end date cannot be before the start date.")`.
4. `to.DayNumber - from.DayNumber >= 3660` (`MaxRangeDays`) → `AnalyticsFilterException("The date range is too long.")`.
5. `!Enum.IsDefined(filter.Basis)` → `AnalyticsFilterException("The date basis is not valid.")`.
6. With `checkEntitiesExist` (Administrator): each given department / project / employee / manager id must exist (`DepartmentExistsAsync`, `ProjectExistsAsync`, `UserExistsAsync`), else `AnalyticsEntityNotFoundException("The {kind} does not exist.")`.
7. Return `ResolvedAnalyticsFilter` with `FromUtc` = local midnight of `from`, `ToUtcExclusive` = local midnight of `to + 1`, the ids, `AllEmployees`, `Today`, `NowUtc`, `TimeZone`.

Service logic – `AdminAnalyticsService.GetAsync(filter)` (FR-DASH-16 – FR-DASH-39)

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1. `AdminDashboardScope.ScopeAsync` → `ScopedTasks(Tasks, TasksIgnoringDates, Filter)`.
2. `CountByStatusAsync(scoped.Tasks, now)` → `BucketCountsDto` (also the "Total tasks … Completion" cards).
3. Summary: `CountEmployeesAsync(PermissionKeys.TasksViewAll, employeeId, departmentId, managerId)` (active users whose role does not hold `tasks.view.all`), `CountManagersAsync` (`RoleId == SystemRoles.ManagerId`), `CountDepartmentsAsync`, `CountProjectsAsync`.
4. `CountByPriorityAsync` (every `TaskPriority`, zero-filled) and `CountByDeadlineAsync` with `DeadlineBucketRules.BoundsFor(now, timeZone)`.
5. `TrendAsync(scoped.TasksIgnoringDates)`: created and completed instants per quarter hour in [`FromUtc`, `ToUtcExclusive`) (`completedStatusOnly: false`), grouped into local days; weekly buckets starting Monday (`StartOfWeek`) when the range is longer than `DailyTrendMaxDays` (31); `Granularity` = `day` or `week`.
6. `ByDepartmentAsync`: `CountByDepartmentAsync` joined to `GetDepartmentsAsync(departmentId)`; unmatched keys summed into a `No department` row when its total is above 0.
7. `ByEmployeeAsync`: `CountByPersonAsync`, ordered by open, total, id; `Take(TopEmployees = 10)` unless `AllEmployees`; names from `GetUserNamesIncludingDeletedAsync` (fallback `Unknown`); `TotalEmployees` = number of people with tasks.
8. `ByManagerAsync`: `GetManagersAsync(managerId)` with `CountByDepartmentManagerAsync`, `GetDepartmentTeamSizesAsync`, `GetManagedDepartmentCountsAsync`, `GetManagedProjectCountsAsync`.
9. `ByProjectAsync`: `CountByProjectAsync` + `GetLatestDeadlineByProjectAsync` (`ProjectedEnd`) joined to `GetProjectsAsync(projectId, null)`; `No project` row for the rest.
10. `GetFilterOptionsAsync`: `today`, `timeZone.Id`, all departments, projects, users and Manager-role users by name.

Service logic – `ManagerAnalyticsService.GetAsync(managerId, filter)` (FR-DASH-40 – FR-DASH-58)

1. `ManagerDashboardScope.ScopeAsync`: `filter with { DepartmentId = null, ManagerId = null }`; `EnsureInScopeAsync` (employee not in `TeamUserIdsAsync` → `AnalyticsForbiddenException("The employee is outside your team.")` if the user exists, else `AnalyticsEntityNotFoundException("The employee does not exist.")`; project not in `ProjectIdsAsync` → `"The project is outside your scope."` / `"The project does not exist."`); resolve without entity checks; set = `AnalyticsTaskSet(TeamUserIds: team, ProjectId, EmployeeId)` + dates.
2. Team = `IEmployeeRepository.GetTeamMemberIdsAsync`: active users other than the manager in a department the manager manages (`DepartmentManagers`) or in a project where the manager's allocation has `IsProjectManager = true`; names via `GetUserNamesAsync`.
3. `CountByStatusAsync`, `CountByPriorityAsync`; `ByProjectAsync` limited to `ProjectIdsAsync` (`ManagerProjects` ∪ projects of team tasks) plus `No project`.
4. `TeamSummaryDto(teamSize, totalProjects, byStatus)`: team size = 1 when an employee filter is set; projects = 1 when a project filter is set.
5. `ByEmployeeAsync`: baseline = whole team (with an employee filter the scope is re-run without the employee, dates pinned to the first resolution, same `NowUtc`); `CountByPersonAsync(baseline)`; mean open over all members; `HighWorkload = ManagerTeamThresholds.IsHighWorkload(open, mean)` (open ≥ 5 and ≥ 1.5 × mean); `HasOverdue = overdue > 0`; rows narrowed to the filtered employee; ordered by open, total, name.
6. `AttentionAsync`: `GetOverdueAsync(take: AttentionListSize = 5)` ordered by deadline then id; assignee and project names including deleted; `DaysOverdue` = whole local days between deadline day and today (never negative); projects needing attention = top 5 projects (not `No project`) by overdue count, then name.

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7. `GetFilterOptionsAsync(managerId)`: team members (ordinal name order) and in-scope projects; no departments or managers.

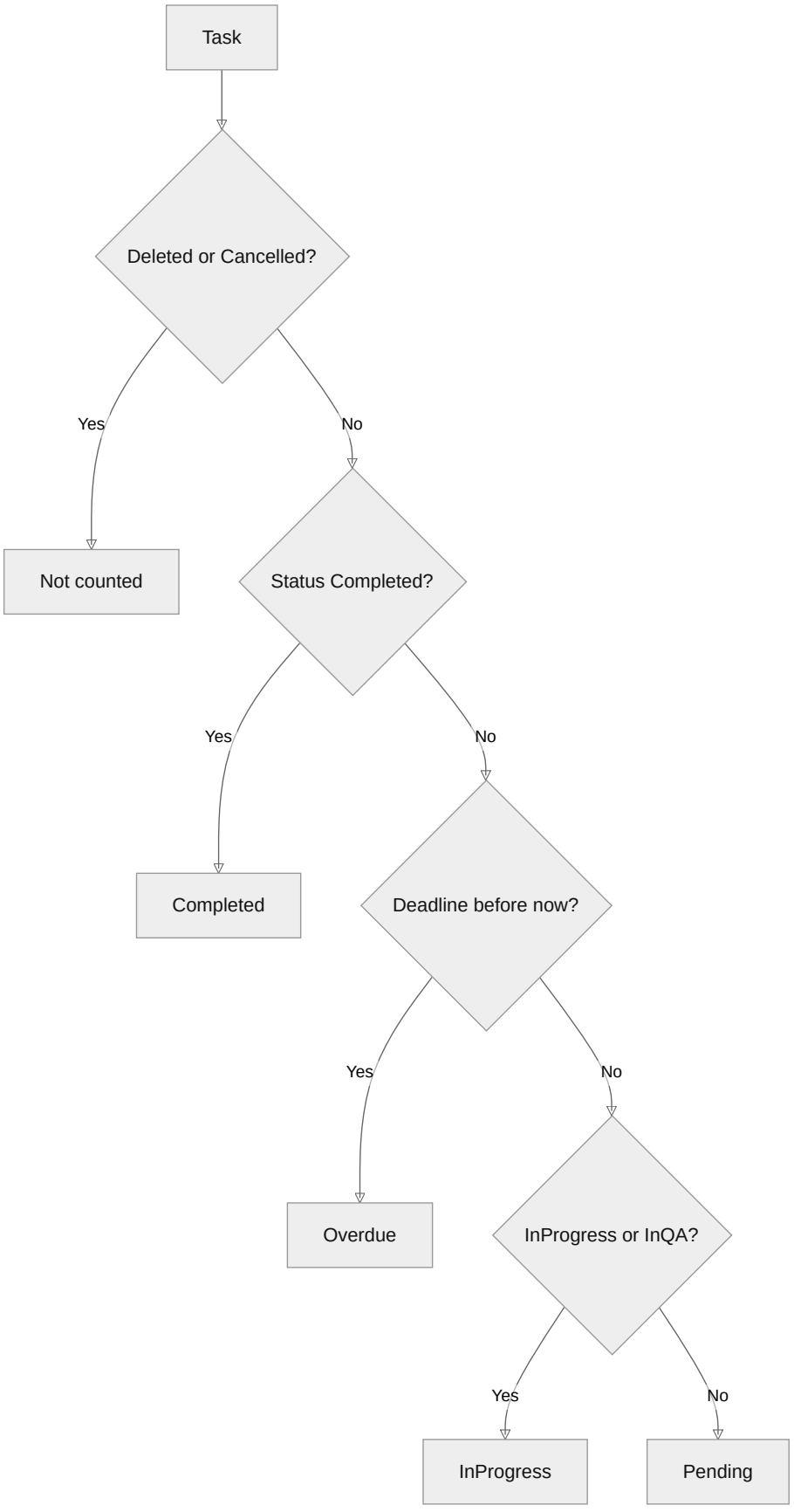
Service logic – `EmployeeAnalyticsService.GetAsync(userId, period)` (FR-DASH-59 – FR-DASH-71)

1. `ParsePeriod`: `null`, `""`, `30d` → `Last30Days`; `7d` → `Last7Days`; `month` → `ThisMonth`; else `AnalyticsFilterException("The period is not valid.")`.
2. `from` = today − 6, today − 29 or the 1st of the month; `to` = today.
3. `own = EmployeeDashboardScope.OwnTasks(userId)` – the id always comes from the session claim; no other query key is read.
4. Status, priority, deadlines, projects and department use **all** own tasks; only the trend and `CompletedInPeriod` use the period. Trend: per local day, completed with `completedStatusOnly: true`.
5. `UpcomingAsync`: `GetOpenDueBeforeAsync(own, horizon, UpcomingListSize = 10)` with the 7-day horizon from `DeadlineBucketRules.BoundsFor`; `DaysRemaining` (local days), `IsOverdue` (`Deadline < now`), `DaysOverdue`.
6. `ProjectsAsync`: `GetProjectProgressAsync(own)` ∪ `GetAllocatedProjectIdsAsync(userId)`; `GetProjectsByIdsAsync`; by name; own total, completed, `ProgressPercent`, `LatestDeadline` (latest open deadline).
7. `DepartmentAsync`: `GetFirstDepartmentIdAsync` (first by name) → null if none; if `CountActiveDepartmentMembersAsync < MinDepartmentMembersForBreakdown (3)` → `EmployeeDepartmentDto(name, true, null, null)`; else `CountByStatusAsync(AnalyticsTaskSet(ActiveMembersOfDepartmentId: departmentId))`. The DTO has no per-user field.

Bucket rule – `TaskBucketRules` (FR-DASH-10 – FR-DASH-12)

- `IsCounted()` = `!IsDeleted && Status != Cancelled`.
- `BucketOf(now)`: Completed if `Status == Completed`; else Overdue if `Deadline < now`; else InProgress if `InProgress` or `InQA`; else Pending.
- `CompletionPercent(completed, total)` = 0 when total ≤ 0, else `Math.Round(completed × 100 / total, MidpointRounding.AwayFromZero)`.
- `DeadlineBucketRules.InBucket` (open = not Completed, not Cancelled): Overdue `Deadline < now`; DueToday `now ≤ Deadline < tomorrow`; Next7Days `tomorrow ≤ Deadline < horizon` (tomorrow + 7 days); Later `Deadline ≥ horizon`.

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Frontend logic

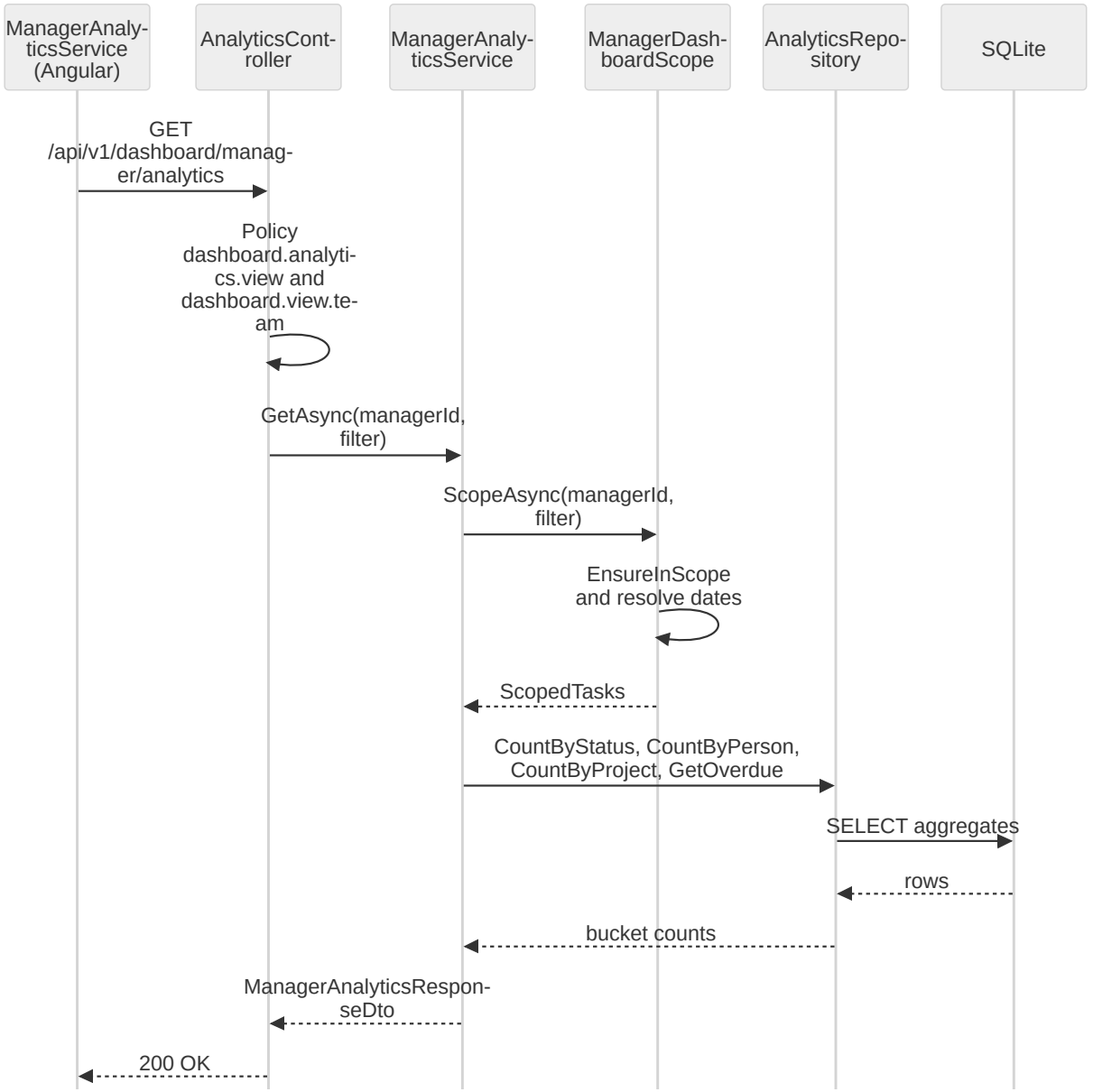
1. Each page calls `start()` on its service in `ngOnInit`: Administrator / Manager load the options (`.../filters`, only when `canFilter()` = holds `dashboard.filters.use`) and the analytics with

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`defaultFilter(localToday())` (Last 30 days, `Created`), replacing "today" by the server `today` from the options.

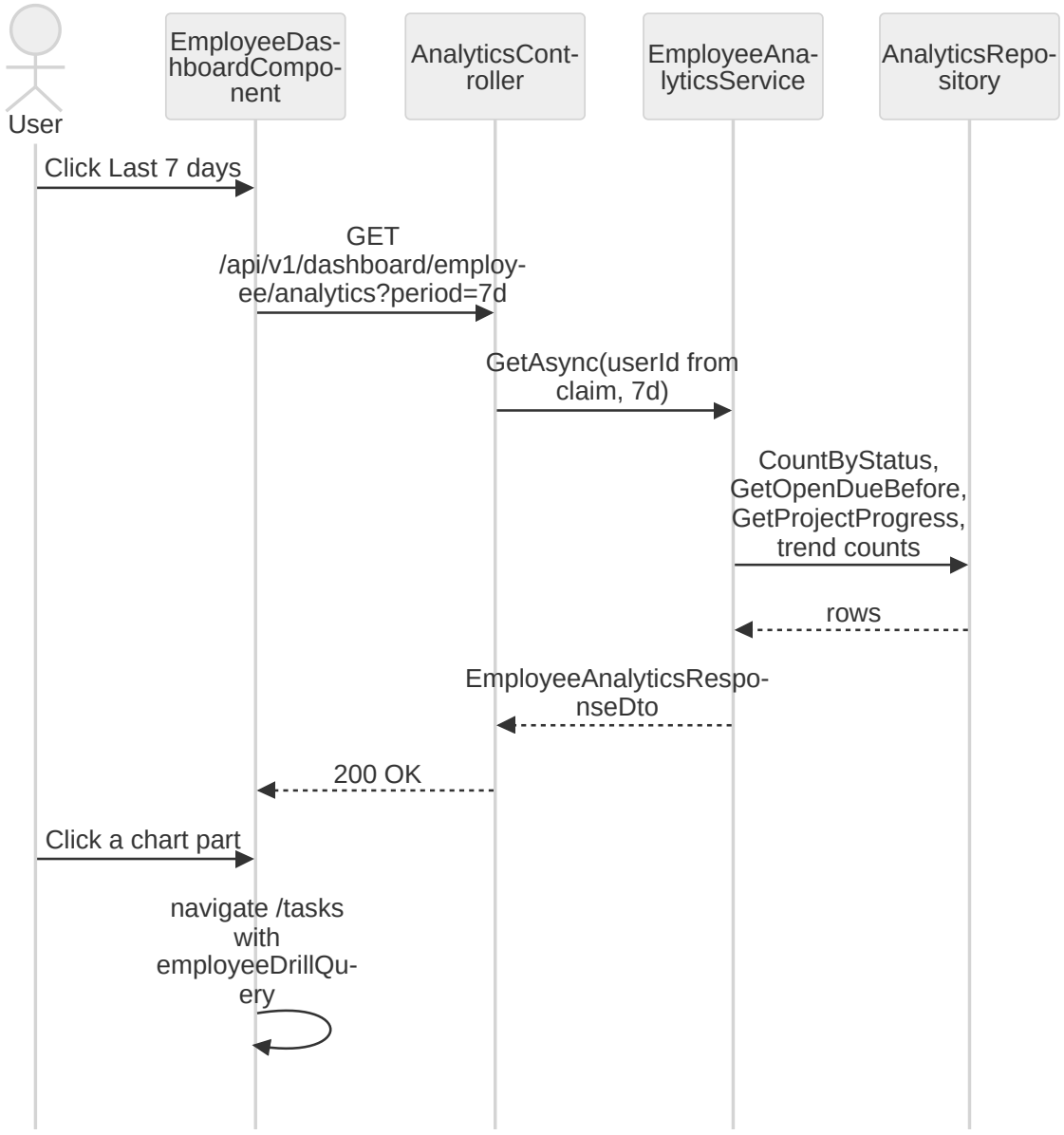
2. Filter changes go through `update(patch)`; `rangeError` = `validateRange(from, to)` ("Choose both a start and an end date." / "The end date cannot be before the start date."); an invalid range sends no request. `reset()` restores `defaultFilter`. `switchMap` cancels stale responses.
3. Any HTTP error sets `error` to "Could not load analytics. Check your connection and try again." (Administrator / Manager) or "Could not load your dashboard. Check your connection and try again." (Employee); `retry()` calls `refresh()`.
4. Drill-down (FR-DASH-72 – FR-DASH-75): `openTasks(target)` navigates to `/tasks` with `drillDownQuery(filter, target)` (Administrator / Manager: `from`, `to`, `basis`, ids, `bucket` or `deadlineBucket`, `priority`; `0` for "No department" / "No project") or `employeeDrillQuery(userId, target)` (Employee). The Task List parses it with `parseDrillDown` and its repository reuses `TaskBucketRules` / `DeadlineBucketRules` and the caller's task access (Task Management LLD).

Sequence diagram – Manager analytics



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Sequence diagram – Employee analytics and drill-down



Repository methods – `AnalyticsRepository` (`IAnalyticsRepository`). Every task query starts from `Build(AnalyticsTaskSet)`: `AsNoTracking`, `TaskBucketRules.IsCounted()`, team / own / active-department-member filters, `DashboardTaskFilters.ApplyEntityFilters`, then `ApplyDateRange` on `CreatedAt`, `Deadline` or `CompletedAt` (`CompletedAt != null` for the Completed basis), end exclusive.

Method	Query / purpose	Notes
`CountByStatusAsync(set, now)`	Pending / InProgress / Completed / Overdue	`AnalyticsCounting.TotalsAsync`: one `COUNT` per `TaskBucket` with `TaskBucketRules.InBucket`
`CountByPriorityAsync(set)`	Count per priority	`GROUP BY Priority`, zero-filled for every `TaskPriority`
`CountByDeadlineAsync(set, bounds)`	Overdue / DueToday / Next7Days / Later	Open tasks only (`DeadlineBucketRules.InBucket`)

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Method	Query / purpose	Notes
`CountCreatedPerQuarterHourAsync`, `CountCompletedPerQuarterHourAsync(…, completedStatusOnly)`	Instants per 15-minute UTC slot	`AnalyticsCounting.QuarterHourCountsAsync` groups by year/month/day/hour/quarter; the service groups into local days
`CountByDepartmentAsync(set, now)`	Buckets per department of the assignee	Over `UserDepartments`; a person in several departments counts in each; key `0` = no department
`CountByPersonAsync(set, now)`	Buckets per person	Union of assignee pairs and `TaskRecipients` pairs
`CountByDepartmentManagerAsync(set, now)`	Buckets per department manager	`DepartmentManagers` × `UserDepartments`
`CountByProjectAsync(set, now)`, `GetLatestDeadlineByProjectAsync(set)`	Buckets / latest deadline per project	Key `0` = no project
`GetOverdueAsync(set, now, take)`	Most overdue tasks (`OverdueTaskRow`)	`ORDER BY Deadline, Id`, `TAKE`
`GetOpenDueBeforeAsync(set, beforeUtc, take)`	Open tasks due before a horizon (`DeadlineTaskRow`)	Status not Completed; soonest first
`GetProjectProgressAsync(set)`	Per project total, completed, latest open deadline (`ProjectProgressRow`)	Tasks with a project only
`GetDepartmentsAsync`, `GetProjectsAsync(only, within)`, `GetUsersAsync`, `GetManagersAsync`	Option lists (`AnalyticsOptionDto`)	Ordered by name; live rows
`CountEmployeesAsync`, `CountManagersAsync`, `CountDepartmentsAsync`, `CountProjectsAsync`	Summary counts	`CountEmployeesAsync` excludes roles holding `tasks.view.all` via `RolePermissions` / `Permissions`
`GetDepartmentTeamSizesAsync`, `GetManagedDepartmentCountsAsync`, `GetManagedProjectCountsAsync`	Per-manager sizes	`DepartmentManagers`, `ManagerProjects`
`GetUserNamesAsync`, `GetUserNamesIncludingDeletedAsync`, `GetProjectNamesIncludingDeletedAsync`	Name lookups	`IgnoreQueryFilters` for the "including deleted" variants
`DepartmentExistsAsync`, `ProjectExistsAsync`, `UserExistsAsync`	Existence checks	Respect soft-delete filters
`GetManagerProjectIdsAsync(managerId, teamUserIds)`	Manager's project scope	`ManagerProjects` ∪ projects of team tasks

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Method	Query / purpose	Notes
`GetFirstDepartmentIdAsync`, `GetDepartmentNameAsync`, `CountActiveDepartmentMembersAsync`, `GetAllocatedProjectIdsAsync`, `GetProjectsByIdsAsync`	Employee helpers	–
`GetTaskIdsAsync`, `CountTasksAsync`	Ids / count of a set	On the interface; no caller found outside the repository

`DashboardTaskFilters.ApplyEntityFilters`: department `0` = assignee in no department, else assignee is a member (`IgnoreQueryFilters` on `Users`); project `0` = no project (or a deleted one), else `ProjectId == id`; employee = assignee or recipient; manager = assignee in a department the manager manages (`TeamUserIds`, department membership only).

Validation rules

Field	Rule	Message
`from` / `to` (client)	Both required	"Choose both a start and an end date." (no request sent)
`from` / `to`	`to` not before `from`	"The end date cannot be before the start date."
Range	`to - from` < 3660 days	"The date range is too long."
`basis`	Defined `AnalyticsDateBasis` value (an unparseable name is refused by model binding)	"The date basis is not valid."
`departmentId` / `projectId` / `employeeId` / `managerId` (Administrator)	Must exist	"The department does not exist." / "The project does not exist." / "The employee does not exist." / "The manager does not exist."
`employeeId` (Manager)	Must be a team member	"The employee is outside your team."
`projectId` (Manager)	Must be in `ProjectIdsAsync`	"The project is outside your scope."
`period` (Employee)	`7d`, `30d`, `month`, or empty	"The period is not valid."

Error cases

Case	Exception	Status
Reversed range, range too long, bad basis, bad period	`AnalyticsFilterException`	400
Employee / project outside the manager's scope	`AnalyticsForbiddenException`	403
Named id does not exist	`AnalyticsEntityNotFoundException`	404
Missing `dashboard.analytics.view` / `dashboard.filters.use` / view key	Policy	403
Caller id not numeric (manager, employee)	–	401

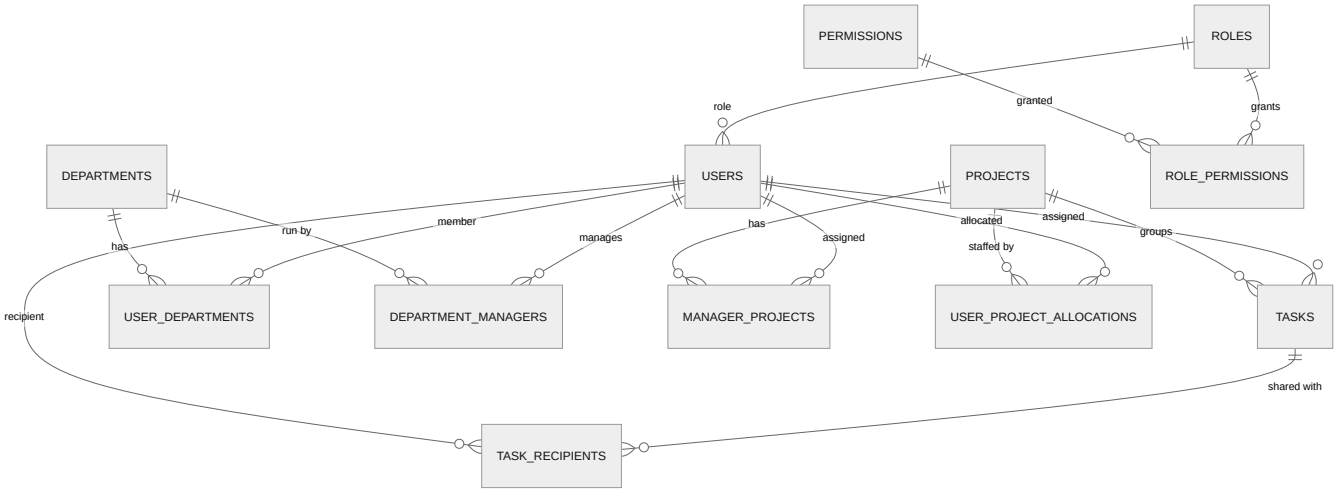
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State machine: *Not applicable for this child module* – it reads task status; transitions are owned by Task Management. The bucket classification is shown as a decision flow above.

5. Database Design

The Dashboard owns **no table** and runs **no migration**. It reads the tables below; their full definition is owned by the Task Management, Employee, Department and Project Management LLDs and the overall LLD. Only the columns this sub-module reads are listed.

5.1 ER Diagram



5.2 Tables

`Tasks` (Task)

Column	Type	Null	Key / index	Default	Description
Id	int	No	PK	identity	Task id; drill-down link `/tasks/:id`
TaskCode	string(32)	No	Unique `IX_Tasks_TaskCode`	–	Shown in deadline / overdue lists
Title	string(200)	No	–	–	Shown in lists
AssignedToUserId	int	No	FK → `Users.Id` (Restrict), `IX_Tasks_AssignedToUserId`	–	Owner (with recipients); person, team, department grouping
ProjectId	int	Yes	FK → `Projects.Id` (Restrict), `IX_Tasks_ProjectId`	null	Project grouping; null = "No project"
Priority	`TaskPriority` (int)	No	–	–	Priority donut
Status	`TaskStatus` (int)	No	–	–	Buckets

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Column	Type	Null	Key / index	Default	Description
Deadline	DateTime (UTC)	No	`IX_Tasks_Deadline`	–	Overdue, deadline windows, Deadline basis
CreatedAt	DateTime (UTC)	No	–	–	Created basis, trend
CompletedAt	DateTime (UTC)	Yes	–	null	Completed basis, trend
IsDeleted	bool	No	–	false	Deleted tasks are never counted

`TaskRecipients` (TaskRecipient)

Column	Type	Null	Key / index	Default	Description
Id	int	No	PK	identity	–
TaskId	int	No	FK → `Tasks.Id` (Cascade); unique `IX_TaskRecipients_TaskId_UserId` (TaskId, UserId)	–	A recipient owns the task for team / own figures
UserId	int	No	FK → `Users.Id` (Cascade), `IX_TaskRecipients_UserId`	–	Recipient

`Users` (User)

Column	Type	Null	Key / index	Default	Description
Id	int	No	PK	identity	Person
FullName	string(200)	No	–	–	Names in lists and options
RoleId	int	No	FK → `Roles.Id` (Restrict), `IX_Users_RoleId`	–	Managers = `SystemRoles.ManagerId` (2); Employees card excludes roles with `tasks.view.all`
IsActive	bool	No	–	–	Team and department members must be active
IsDeleted	bool	No	Query filter `!IsDeleted`	false	Deleted names still shown via `IgnoreQueryFilters`

`Departments` (Department), `Projects` (Project)

Column	Type	Null	Key / index	Default	Description
Id	int	No	PK	identity	–
Name	string(200)	No	Unique among live rows (`IX_Departments_Name` / `IX_Projects_Name` , filter `"IsDeleted" = 0`)	–	Labels, ordering
IsDeleted	bool	No	Query filter	false	Soft delete

`UserDepartments` (UserDepartment), `DepartmentManagers` (DepartmentManager), `ManagerProjects` (ManagerProject)

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Table	Columns	Key / index	Description
`UserDepartments`	UserId, DepartmentId (int)	Composite PK; FKs Cascade; `IX_UserDepartments_DepartmentId`; query filter user and department not deleted	Department membership (team, department rows, department overview)
`DepartmentManagers`	ManagerUserId, DepartmentId (int)	Composite PK; FKs Cascade; `IX_DepartmentManagers_DepartmentId`	Departments a manager runs
`ManagerProjects`	ManagerUserId, ProjectId (int)	Composite PK; FKs Cascade; `IX_ManagerProjects_ProjectId`	Projects assigned to a manager (project scope, Projects column)

`UserProjectAllocations` (UserProjectAllocation)

Column	Type	Null	Key / index	Default	Description
UserId	int	No	Composite PK; FK → `Users.Id` (Cascade)	–	Member
ProjectId	int	No	Composite PK; FK → `Projects.Id` (Cascade), `IX_UserProjectAllocations_ProjectId`	–	Project
IsProjectManager	bool	No	–	false	Widens a manager's team to the project's members
AllocationPercent, PositionId	int	No	Check 1–100; FK → `Positions` (Restrict)	–	Not read by the Dashboard

`Roles` (Role), `RolePermissions` (RolePermission), `Permissions` (Permission)

Read only by `CountEmployeesAsync` to exclude roles whose `RolePermissions` include the `Permissions.Key` `tasks.view.all` (composite PK `RoleId`, `PermissionId`; unique `Permissions.Key`).

5.3 Indexes, Constraints & Triggers

Object	Definition	Purpose
`IX_Tasks_AssignedToUserId`	(AssignedToUserId)	Team / own / person filters
`IX_Tasks_Deadline`	(Deadline)	Overdue and deadline windows, Deadline basis, ordering of lists
`IX_TaskRecipients_TaskId_UserId`	Unique (TaskId, UserId)	Recipient ownership lookups
Soft-delete query filters	`Users`, `Departments`, `Projects`, `UserDepartments`, `UserProjectAllocations`	Live rows only, unless `IgnoreQueryFilters`

No triggers are used.

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5.4 Enumerations

Enum	Values	Stored as
`TaskStatus`	ToDo (0), InProgress (1), Blocked (2), Completed (3), Cancelled (4), InQA (5)	int; JSON name
`TaskPriority`	Low (0), Medium (1), High (2), Critical (3)	int; JSON name
`AnalyticsDateBasis`	Created, Deadline, Completed	Not stored; query parameter (name)
`EmployeeAnalyticsPeriod`	Last7Days, Last30Days, ThisMonth	Not stored; response field (query uses `7d`, `30d`, `month`)
`TaskBucket`	Pending, InProgress, Completed, Overdue	Not stored; computed
`DeadlineBucket`	Overdue, DueToday, Next7Days, Later	Not stored; computed

5.5 Migrations

Not applicable for this sub-module – it adds no migration. The tables it reads come from `20261012124843_AddTasks`, `20261020173455_RenameDepartmentsToProjectsAndAddDepartments`, `20261023160825_UserProjectAllocationsAndDepartments` and `20261023173211_AddProjectManagerFlag` (described in the owning LLDs).

6. Algorithms & Background Jobs

Job / algorithm	Trigger / schedule	Steps	Idempotency
Bucket classification (`TaskBucketRules`)	Every analytics query	Counted → Completed → Overdue → InProgress → Pending (4.1.2)	Pure function of task and `now`
Deadline windows (`DeadlineBucketRules.BoundsFor`)	Admin deadline chart, Employee deadlines	`now`; tomorrow = local midnight of today + 1; horizon = tomorrow + 7 days, as UTC	Pure
Trend bucketing (`AdminAnalyticsService.TrendAsync`, `EmployeeAnalyticsService.TrendAsync`)	Admin / Employee analytics	Quarter-hour UTC counts → local day (or Monday week when > 31 days) → zero-filled points	Pure
High-workload flag (`ManagerTeamThresholds`)	Manager analytics	`open ≥ 5 && open ≥ 1.5 × team mean`; overdue flag `overdue > 0`	Pure
Background jobs	–	None in this sub-module	–

7. Security Design (detailed)

Shared items (password hashing, tokens, cookies, secrets) are in the overall LLD.

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Item	Implementation
Permission checks	`AnalyticsController` actions: `[Authorize(Policy = PermissionPolicy.Prefix + PermissionKeys.DashboardAnalytics / DashboardFilters)]` + `[RequirePermission(DashboardViewOrganization / DashboardViewTeam)]`; `DashboardController`: `RequirePermission` (admin, manager) or `[Authorize]` (employee)
Scope expansion	`EffectivePermissions.Expand`: `dashboard.view.organization` implies `dashboard.view.team`; legacy `dashboard.admin` → `dashboard.view.organization`, `dashboard.manager` → `dashboard.view.team`
Catalogue defaults (`PermissionCatalog`)	`dashboard.view.own` AlwaysGranted, not assignable; `dashboard.view.team` GrantedToManager; `dashboard.view.organization` Administrator only; `dashboard.analytics.view` and `dashboard.filters.use` granted to Manager, Employee and all existing roles
Data scope	Organisation – `AdminDashboardScope`; Team – `ManagerDashboardScope` (403 outside scope); Own – `EmployeeDashboardScope` (session id only; no subject parameter)
Privacy	`EmployeeDepartmentDto` has no per-user field; departments with fewer than 3 active members return `Limited = true` and null figures
Frontend	`permissionGuard` on `/admin`, `/manager`; `canFilter()` hides the filter bar; no directive on cards
Audit	`DeniedActionAuditHandler` logs refused policy checks

8. Configuration

Key	Default	Purpose
`Application:Timezone`	`Asia/Kolkata`	Fallback application timezone (overridden by the stored application setting via `IAppClock`) for "today" and day bounds
`ConnectionStrings:Default`	`<secret>`	SQLite database
`Jwt:AccessTokenMinutes` / `Jwt:RefreshTokenDays`	`<secret>`	Session lifetime for dashboard calls

Code constants (not configurable): `AnalyticsFilterResolver.DefaultRangeDays` 30, `MaxRangeDays` 3660; `AdminAnalyticsService.DailyTrendMaxDays` 31, `TopEmployees` 10; `ManagerTeamThresholds.AttentionListSize` 5, `HighWorkloadFactor` 1.5, `HighWorkloadMinimumOpen` 5; `EmployeeAnalyticsService.UpcomingListSize` 10, `MinDepartmentMembersForBreakdown` 3; `DeadlineBucketRules.UpcomingWindowDays` 7; `DashboardService.UpcomingDeadlineWindowDays` 7.

9. Testing Design

Paths relative to `backend/tests`. Only file names are available; "Covers" is inferred from the names.

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Area	Test class / file	Type	Covers
Shared rules	`TaskFlow.UnitTests/Common/TaskBucketRulesTests.cs`	Unit	Bucket classification, completion percent
Overview	`TaskFlow.UnitTests/Dashboard/Overview/DashboardServiceTests.cs` (+ `DashboardServiceHarness.cs`)	Unit	Counter endpoints' service
Overview	`TaskFlow.IntegrationTests/Dashboard/Overview/DashboardEndpointsTests.cs`	Integration	Counter endpoints and authorization
Admin analytics	`TaskFlow.UnitTests/Dashboard/Analytics/Admin/AdminAnalyticsServiceTests.cs`, `AdminAnalyticsQueryShapeTests.cs`	Unit	Cards, charts, breakdowns, query shape
Manager analytics	`TaskFlow.UnitTests/Dashboard/Analytics/Manager/ManagerAnalyticsServiceTests.cs`	Unit	Team figures, flags, attention
Employee analytics	`TaskFlow.UnitTests/Dashboard/Analytics/Employee/EmployeeAnalyticsServiceTests.cs`	Unit	Periods, deadlines, projects, department privacy
Scopes	`TaskFlow.UnitTests/Dashboard/Analytics/Scopes/AnalyticsScopeTests.cs`, `ManagerScopeTests.cs`	Unit	Filter resolution, team / project scope
Drill-down	`TaskFlow.UnitTests/Dashboard/Analytics/Scopes/AnalyticsDrillDownTests.cs`, `ManagerDrillDownScopeTests.cs`	Unit	Chart count = Task List count
Support	`TaskFlow.UnitTests/TestSupport/AnalyticsHarness.cs`	Support	Shared harness
Endpoints	`TaskFlow.IntegrationTests/Dashboard/Analytics/AdminAnalyticsEndpointsTests.cs`, `ManagerAnalyticsEndpointsTests.cs`, `EmployeeAnalyticsEndpointsTests.cs`	Integration	Endpoints, policies, validation responses
Isolation	`TaskFlow.IntegrationTests/Dashboard/Analytics/ManagerAnalyticsIsolationTests.cs`, `EmployeeAnalyticsIsolationTests.cs`	Integration	Team and own-data isolation
Frontend	–	–	No frontend spec file is in the snapshot

10. Known Limitations & Open Issues

- The Overview counter endpoints and the Angular `DashboardService.getEmployeeStats()` are not used by any screen; their counting differs from Analytics (Cancelled counted in Total; "pending" = all open tasks).
- Team definition differs: Manager Dashboard team (`GetTeamMemberIdsAsync`: managed departments + projects where the manager is flagged project manager, active only) vs. Administrator Manager filter and "Manager-wise teams" (`DashboardTaskFilters.TeamUserIds` / `CountByDepartmentManagerAsync`: department membership only). > ⚠️ To be confirmed: one team definition (R-DASH-03).
- Trend completion rule differs: Administrator `completedStatusOnly: false`, Employee `true`. > ⚠️ To be confirmed: one rule (R-DASH-04).

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- The UI does not check `dashboard.analytics.view`; a role without it sees only the load-error banner. > ⚠️ To be confirmed: a "no permission" message (R-DASH-01).
- The Angular services ignore the server `detail`, so 400 / 403 / 404 look like a connection problem. > ⚠️ To be confirmed (R-DASH-02).
- `GET /api/v1/dashboard/employee/analytics` stacks a redundant `[Authorize]`; the period is not gated by `dashboard.filters.use` (by design, BR-DASH-08).
- The "Managers" card counts only the Manager system role (`RoleId == 2`); custom roles with `dashboard.view.team` are not counted.
- `GetTaskIdsAsync` / `CountTasksAsync` on `IAnalyticsRepository` have no caller outside the repository.
- Many sequential aggregate queries per request (one per bucket per grouping). > ⚠️ To be confirmed: a measured load-time target (NFR-DASH-05).

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11. Traceability (FRD → code)

FRD requirement	Endpoint / class	Test
FR-DASH-01, FR-DASH-02	`dashboardRouteFor`, `ROLE_ROUTES` (`shared/auth/role-routes.ts`); `nav-items.ts`	Manual UI check (AC-DASH-01)
FR-DASH-03, FR-DASH-04	`DASHBOARD_ROUTES` + `permissionGuard` (`/admin`, `/manager`)	`AnalyticsScopeTests`; manual UI check
FR-DASH-05	`DASHBOARD_ROUTES` `me` (no guard) · `GET /api/v1/dashboard/employee/analytics`	`EmployeeAnalyticsEndpointsTests`
FR-DASH-06	`RequirePermission(dashboard.view.organization / dashboard.view.team)` on endpoints 1, 2, 4 – 7	`AdminAnalyticsEndpointsTests`, `ManagerAnalyticsEndpointsTests`, `DashboardEndpointsTests`
FR-DASH-07	`perm:dashboard.analytics.view` on endpoints 4, 6, 8	`AdminAnalyticsEndpointsTests`, `EmployeeAnalyticsEndpointsTests`
FR-DASH-08, FR-DASH-09	`loading` / `error` signals, `retry()` in the three analytics services and pages	Manual UI check (AC-DASH-03)
FR-DASH-10, FR-DASH-11, FR-DASH-12	`TaskBucketRules.IsCounted`, `BucketOf`, `CompletionPercent`; `BucketCountsDto`	`TaskBucketRulesTests`, `AdminAnalyticsServiceTests`
FR-DASH-13	`AnalyticsTaskSet.TeamUserIds` / `OwnUserId` (assignee or `TaskRecipients`) · `CountByPersonAsync`	`ManagerAnalyticsServiceTests`, `EmployeeAnalyticsServiceTests`
FR-DASH-14	`IAppClock.GetTimeZoneAsync`, `ReminderCalculator.LocalMidnightUtc`, `DeadlineBucketRules.BoundsFor`	`EmployeeAnalyticsServiceTests`, `AnalyticsScopeTests`
FR-DASH-15	All queries `AsNoTracking`; no `IUnitOfWork` in the sub-module	Manual check – no data change

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FRD requirement	Endpoint / class	Test
FR-DASH-16, FR-DASH-17, FR-DASH-18	`GET /api/v1/dashboard/admin/analytics` · `AdminAnalyticsService.SummaryAsync` (`CountEmployeesAsync`, `CountManagersAsync`)	`AdminAnalyticsServiceTests`
FR-DASH-19	`CountByStatusAsync`, `CountByPriorityAsync` · `TaskAnalyticsComponent`, `DonutChartComponent`	`AdminAnalyticsServiceTests`
FR-DASH-20, FR-DASH-21, FR-DASH-22	`AdminAnalyticsService.TrendAsync` (`DailyTrendMaxDays`, `StartOfWeek`, `TasksIgnoringDates`) · `LineChartComponent`	`AdminAnalyticsServiceTests`
FR-DASH-23	`CountByDeadlineAsync` + `DeadlineBucketRules`	`AdminAnalyticsServiceTests`
FR-DASH-24	`ByDepartmentAsync` / `CountByDepartmentAsync` · `DepartmentAnalyticsComponent`, `StackedBarChartComponent`	`AdminAnalyticsServiceTests`
FR-DASH-25, FR-DASH-26	`ByEmployeeAsync` (`TopEmployees`, `AllEmployees`) · `DashboardAnalyticsService.setAllEmployees`, `EmployeeAnalyticsComponent`	`AdminAnalyticsServiceTests`; manual UI check
FR-DASH-27	`ByManagerAsync` · `ManagerAnalyticsComponent`	`AdminAnalyticsServiceTests`
FR-DASH-28	`ByProjectAsync` (`ProjectedEnd`) · `ProjectAnalyticsComponent`, `ProgressListComponent`	`AdminAnalyticsServiceTests`
FR-DASH-29	Empty-state templates in `admin-dashboard.component.html` and analytics components (`isEmpty`)	Manual UI check
FR-DASH-30, FR-DASH-31	`canFilter()` · `GET /api/v1/dashboard/admin/analytics/filters` (`perm:dashboard.filters.use`) · `GetFilterOptionsAsync`	`AdminAnalyticsEndpointsTests`
FR-DASH-32, FR-DASH-33, FR-DASH-39	`defaultFilter`, `RANGE_PRESETS`, `presetRange`, `reset()` · `DashboardFilterBarComponent`	Manual UI check
FR-DASH-34	`DashboardTaskFilters.ApplyDateRange` (`AnalyticsDateBasis`)	`AdminAnalyticsServiceTests`
FR-DASH-35	`DashboardTaskFilters.ApplyEntityFilters`	`AdminAnalyticsServiceTests`
FR-DASH-36	`validateRange` / `rangeError`	Manual UI check
FR-DASH-37	`AnalyticsFilterResolver.ResolveAsync` (`MaxRangeDays`, basis, `EnsureExistsAsync`)	`AnalyticsScopeTests`, `AdminAnalyticsEndpointsTests`
FR-DASH-38	`DashboardAnalyticsService.update` + `switchMap`; `rangeLabel`, `basisLabel`	Manual UI check
FR-DASH-40, FR-DASH-49, FR-DASH-57	`GET /api/v1/dashboard/manager/analytics` · `ManagerAnalyticsService.GetAsync` (`TeamSummaryDto`, `ByProjectAsync`) · `TeamTaskAnalyticsComponent`	`ManagerAnalyticsServiceTests`

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FRD requirement	Endpoint / class	Test
FR-DASH-41, FR-DASH-42	`ManagerDashboardScope.TeamUserIdsAsync` → `EmployeeRepository.GetTeamMemberIdsAsync`; `AnalyticsTaskSet.TeamUserIds`	`ManagerScopeTests`, `ManagerAnalyticsIsolationTests`
FR-DASH-43, FR-DASH-44	`noTeam`, `isAdministrator` in `ManagerDashboardComponent` (`teamSize == 0`)	`ManagerAnalyticsServiceTests`; manual UI check
FR-DASH-45, FR-DASH-46	`AttentionAsync` / `GetOverdueAsync` (`DaysOverdue`) · `TeamAttentionComponent` → `/tasks/:id`	`ManagerAnalyticsServiceTests`; manual UI check
FR-DASH-47, FR-DASH-48	`AttentionAsync` projects needing attention · `drillDownQuery`	`ManagerAnalyticsServiceTests`, `ManagerDrillDownScopeTests`
FR-DASH-50, FR-DASH-51, FR-DASH-52, FR-DASH-53	`ManagerAnalyticsService.ByEmployeeAsync`, `ManagerTeamThresholds.IsHighWorkload`, `HasOverdue` · `TeamEmployeeAnalyticsComponent`	`ManagerAnalyticsServiceTests`
FR-DASH-54	`ManagerDashboardScope.ScopeAsync` drops `DepartmentId`, `ManagerId`	`ManagerScopeTests`
FR-DASH-55	`GET /api/v1/dashboard/manager/analytics/filters` · `ManagerAnalyticsService.GetFilterOptionsAsync`	`ManagerAnalyticsEndpointsTests`
FR-DASH-56	`ManagerDashboardScope.EnsureInScopeAsync` → `AnalyticsForbiddenException`	`ManagerScopeTests`, `ManagerAnalyticsIsolationTests`
FR-DASH-58	`EffectivePermissions.Expand` (organization ⇒ team) · `GET /api/v1/dashboard/manager/analytics`	`ManagerAnalyticsEndpointsTests`
FR-DASH-59, FR-DASH-61	`GET /api/v1/dashboard/employee/analytics` · `EmployeeAnalyticsService.GetAsync` (`EmployeeSummaryDto`)	`EmployeeAnalyticsServiceTests`
FR-DASH-60	`EmployeeDashboardScope.OwnTasks(userId)`; controller reads only `period`	`EmployeeAnalyticsIsolationTests`
FR-DASH-62, FR-DASH-63, FR-DASH-64	`UpcomingAsync` / `GetOpenDueBeforeAsync` · `deadlineLabel()` in `EmployeeDashboardComponent` → `/tasks/:id`	`EmployeeAnalyticsServiceTests`; manual UI check
FR-DASH-65	`ParsePeriod`, `EMPLOYEE_PERIODS`, `EmployeeAnalyticsService.setPeriod` (Angular)	`EmployeeAnalyticsEndpointsTests`
FR-DASH-66, FR-DASH-67	`EmployeeAnalyticsService.TrendAsync`, `CompletedInPeriod`	`EmployeeAnalyticsServiceTests`
FR-DASH-68	`ProjectsAsync` (`GetProjectProgressAsync`, `GetAllocatedProjectIdsAsync`)	`EmployeeAnalyticsServiceTests`

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FRD requirement	Endpoint / class	Test
FR-DASH-69, FR-DASH-70, FR-DASH-71	`DepartmentAsync` (`GetFirstDepartmentIdAsync`, `MinDepartmentMembersForBreakdown`), `EmployeeDepartmentDto`	`EmployeeAnalyticsServiceTests`, `EmployeeAnalyticsIsolationTests`
FR-DASH-72, FR-DASH-73	`openTasks()` + `drillDownQuery` / `employeeDrillQuery`; `parseDrillDown` (Task List banner)	`AnalyticsDrillDownTests`
FR-DASH-74, FR-DASH-75	`TaskBucketRules`, `DeadlineBucketRules` reused by the Task List repository; Task List access rules	`AnalyticsDrillDownTests`, `ManagerDrillDownScopeTests`
(no FR – unused counters)	`GET /api/v1/dashboard/admin`, `GET /api/v1/dashboard/manager`, `GET /api/v1/dashboard/employee` · `DashboardService`	`DashboardServiceTests`, `DashboardEndpointsTests`

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